

Chapter 2 · Lesson 1

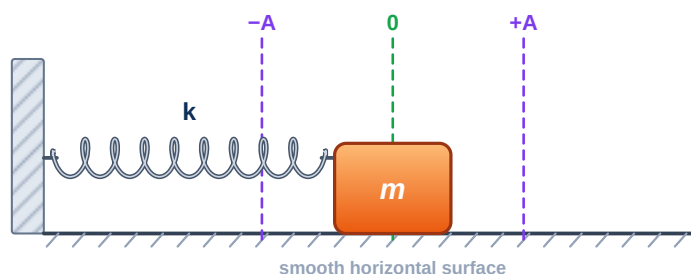
Simple Harmonic Motion

The restoring force · the periodic time and the frequency · the mass-spring oscillator · $x = A \sin \omega t$ · energy in SHM

Take $\pi = 3.14$

1

A block of mass 0.5 kg is attached to a horizontal spring of spring constant 50 N/m on a smooth horizontal surface. The block is pulled aside and released, so it oscillates in simple harmonic motion:



- I The angular frequency of the oscillation equals
- a) 5 rad/s b) 100 rad/s c) 25 rad/s d) 10 rad/s
- II The periodic time of the oscillation equals
- a) 0.628 s b) 0.314 s c) 1.59 s d) 6.28 s

2

The displacement of a body moving in simple harmonic motion is given by the equation $x = 0.05 \sin (20 t)$, where x is in metres and t in seconds:

- I The amplitude of the motion equals
- a) 20 m b) 0.05 m c) 0.1 m d) 1 m
- II The frequency of the motion equals
- a) 20 Hz b) 0.314 Hz c) 3.18 Hz d) 126 Hz

3

Choose the correct answer:

- I** In simple harmonic motion, the restoring force is always
- a) in the direction of motion
 - b) directed towards the equilibrium position and proportional to the displacement
 - c) constant in magnitude
 - d) proportional to the velocity

- II** The speed of the body is maximum at
- a) the equilibrium position
 - b) the two extreme positions
 - c) halfway between them
 - d) every point equally

- III** The acceleration of the body is maximum at
- a) the equilibrium position
 - b) halfway to the extremes
 - c) the two extreme positions
 - d) no point

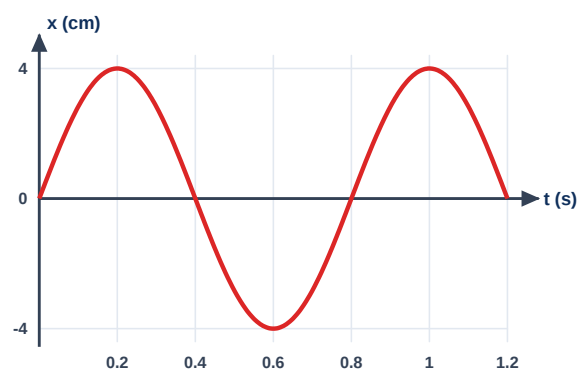
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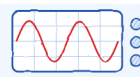
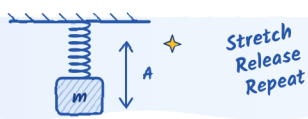
The graph shows how the displacement of a body in simple harmonic motion changes with time:

- I** The amplitude of the motion is
- a) 8 cm
 - b) 4 cm
 - c) 2 cm
 - d) 0.8 cm

- II** The frequency of the motion is
- a) 0.8 Hz
 - b) 2.5 Hz
 - c) 1.25 Hz
 - d) 7.85 Hz

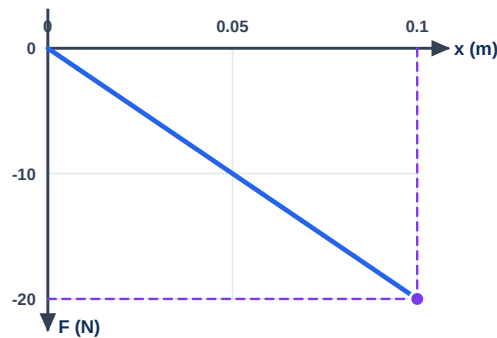
- III** Calculate the angular frequency.





5

The graph shows the restoring force F acting on a body attached to a spring against its displacement x from the equilibrium position:



I The spring constant equals

- a) 2 N/m b) 20 N/m
c) 200 N/m d) 2000 N/m

II If the mass of the body is 2 kg, its periodic time is

- a) 0.628 s b) 0.314 s
c) 6.28 s d) 0.1 s

III What does the negative slope of the line mean?

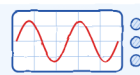
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A body in simple harmonic motion has a maximum speed of 2 m/s and a maximum acceleration of 40 m/s^2 :

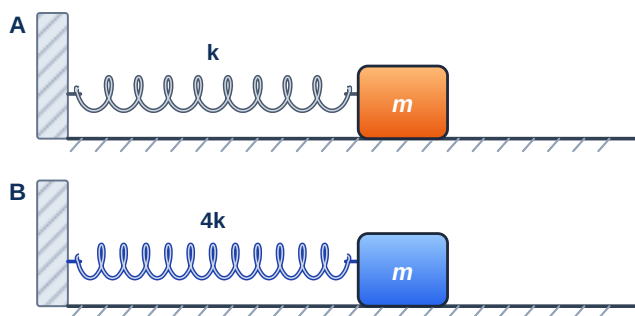
I Its angular frequency equals

- a) 80 rad/s b) 20 rad/s c) 0.05 rad/s d) 40 rad/s

II Find the amplitude and the periodic time.



- 7** Two identical bodies, each of mass m , oscillate horizontally on two springs: spring A of constant k and spring B of constant $4k$, as shown:



- I** The ratio of the periodic times T_A / T_B equals
- a) 4 b) $1/2$ c) $1/4$ d) 2
- II** What mass must be fixed to spring B so that its periodic time equals that of A?

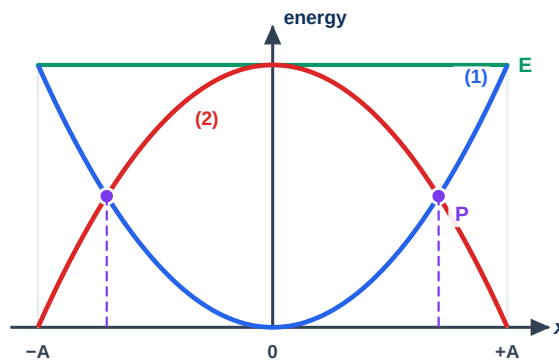
- 8** A body of mass 0.2 kg oscillates on a horizontal spring of constant 80 N/m with an amplitude of 10 cm :

- I** The total mechanical energy of the oscillator is
- a) 0.8 J b) 0.4 J c) 4 J d) 0.04 J
- II** Calculate the maximum speed of the body.

- III** Calculate its speed when the displacement is 5 cm .



The graph shows two energy curves (1) and (2) of a body in simple harmonic motion against its displacement x , together with the total energy E :



- I The curve that represents the kinetic energy is

a) curve (1)
b) none of them

c) the line E
d) curve (2)

- II At point P the kinetic and potential energies are equal. Its displacement is

a) $A/2$
b) $A/4$

c) $A/\sqrt{2}$
d) A

- III Find the ratio KE : PE when $x = A/2$.

10

A body oscillates in simple harmonic motion with a periodic time of 1.2 s, starting from the equilibrium position ($x = A \sin \omega t$):

- I The shortest time taken to move from $x = 0$ to $x = A/2$ is

a) 0.3 s
b) 0.15 s
c) 0.1 s
d) 0.2 s

- II Find the time taken to move from $x = A/2$ to $x = A$, and explain why it is longer than the first half.

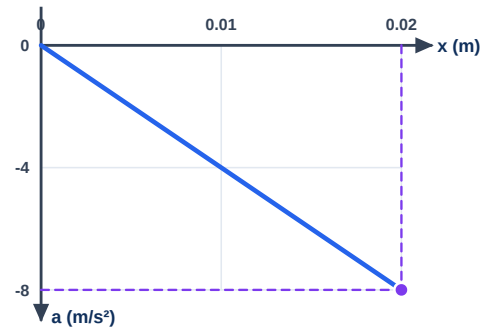
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The graph shows the acceleration a of a body in simple harmonic motion against its displacement x :

I The angular frequency of the motion is

- a) 400 rad/s b) 4 rad/s
c) 0.05 rad/s d) 20 rad/s

II Calculate the periodic time.



III What property of simple harmonic motion does this graph show?

12

The displacement of a body in simple harmonic motion is $x = 0.1 \sin(5\pi t)$, where x is in metres and t in seconds:

I The displacement of the body at $t = 0.1$ s is

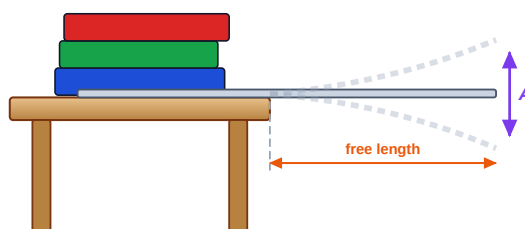
- a) zero b) 0.05 m c) 0.1 m d) -0.1 m

II Find the velocity and the acceleration of the body at that instant.

III Find the periodic time.

13

A steel ruler is clamped at the edge of a table and its free end is set vibrating, as shown:

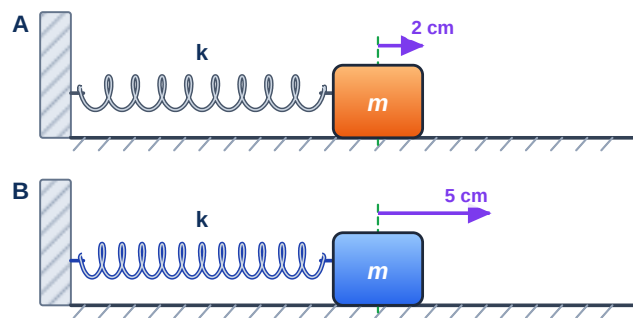


- I** As the amplitude of the vibration slowly decreases, the periodic time
- a) increases b) decreases c) remains constant d) becomes zero

- II** If the free length of the ruler is made shorter, what happens to the frequency? Explain.

14

Two identical bodies on two identical springs are displaced from equilibrium by 2 cm (A) and 5 cm (B) and released at the same moment, as shown:



- I** The body that reaches the equilibrium position first is
- a) A b) B
c) both at the same time d) it cannot be known

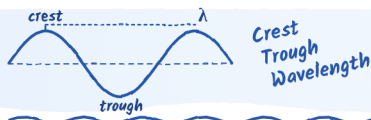
- II** Find the ratio of their maximum speeds $v_A : v_B$.

- III** Find the ratio of their total energies $E_A : E_B$.

15

The amplitude of a mass-spring oscillator is doubled, the mass and the spring being unchanged:

- I** Which of the following is correct?
- a) the periodic time is doubled b) the maximum speed is doubled
c) the total energy is doubled d) the maximum acceleration is quadrupled



I The total energy becomes

- a) 2 times b) the same c) 4 times d) half

16

The spring of a mass-spring oscillator is replaced by one 4 times stiffer, and the mass is doubled:

I The new periodic time T' in terms of the old one T is

- a) $T / \sqrt{2}$ b) $T / 2$ c) $T \sqrt{2}$ d) $2T$

I How does the frequency change?

17

The graph shows the displacement of a body in simple harmonic motion against time. P, Q, R, S and T are five points on the curve:

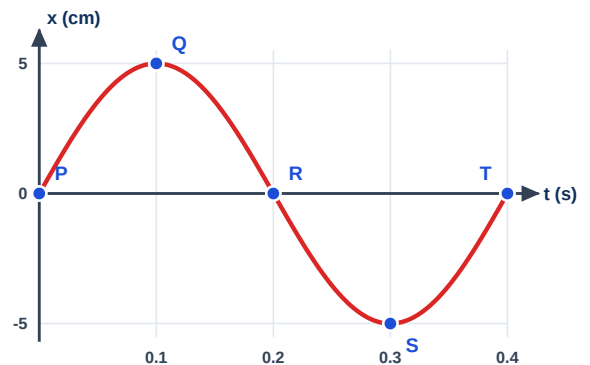
I The speed of the body is maximum at

- a) P, R and T b) Q and S
c) Q only d) S only

I The acceleration is maximum and positive at

- a) Q b) P
c) R d) S

II At which points is the restoring force zero?



18

A body oscillates with an amplitude of 10 cm and an angular frequency of 10 rad/s:

I Its speed when its displacement is 6 cm equals

- a) 1 m/s b) 0.6 m/s c) 0.8 m/s d) 0.36 m/s

I At what displacement is its speed half the maximum speed?

19

The speed of a body in simple harmonic motion is 0.8 m/s when its displacement is 6 cm, and 0.6 m/s when its displacement is 8 cm:

I Find the angular frequency of the motion.

I Find the amplitude.

20

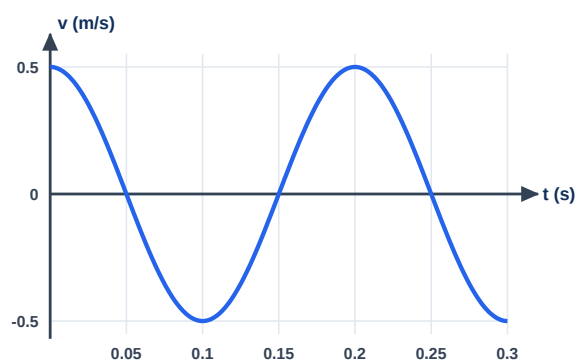
The graph shows how the velocity of a body in simple harmonic motion changes with time:

I Find the angular frequency.

I The amplitude of the motion is

- a) 1.59 cm b) 15.7 cm
c) 0.1 cm d) 3.14 cm

III Where is the body at $t = 0$?



21

Choose the correct answer:

I When the displacement of a body in simple harmonic motion is $A/3$, the ratio KE : PE equals

- a) 3 b) 8 c) 9 d) 1/8

21 A body of mass 0.5 kg oscillates with an amplitude of 4 cm and a frequency of 5 Hz. The maximum restoring force on it is

- a) 19.7 N b) 9.86 N c) 39.4 N d) 0.63 N

22 A student finds that a body of unknown mass m_1 on a spring oscillates with a periodic time of 0.5 s. When a mass of 0.3 kg is added, the periodic time becomes 0.7 s:

I Find the mass m_1 .

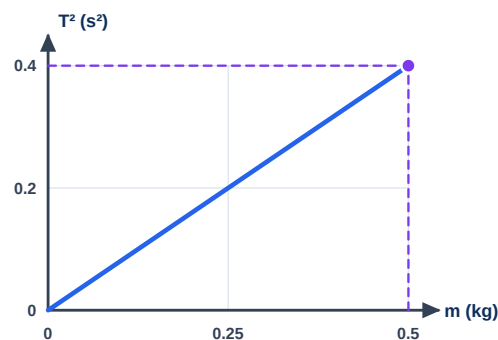
II Find the spring constant.

23 The graph shows the square of the periodic time T^2 of a mass–spring oscillator against the mass m :

I The spring constant equals

- a) 49.3 N/m b) 0.8 N/m
c) 24.6 N/m d) 98.6 N/m

II Find the periodic time when a mass of 0.8 kg is used.

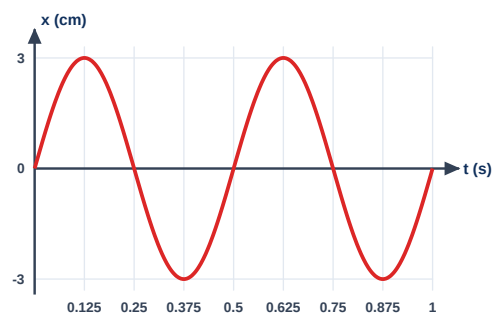


24 The graph shows the displacement–time curve of a body in simple harmonic motion:

I The angular frequency of the motion is

- a) 6.28 rad/s b) 3.14 rad/s
c) 12.56 rad/s d) 2 rad/s

II Write the equation of the displacement in metres.



25

A body starts its simple harmonic motion from the equilibrium position ($x = A \sin \omega t$):

I At $t = T/8$ the ratio KE : PE equals

a) 1 : 1

b) 1 : 3

c) 3 : 1

d) 1 : 2

II A body oscillates with an amplitude of 2.5 cm and a maximum acceleration of 10 m/s^2 . Its frequency is

a) 3.18 Hz

b) 20 Hz

c) 0.314 Hz

d) 400 Hz

26

Give the scientific reason for each of the following:

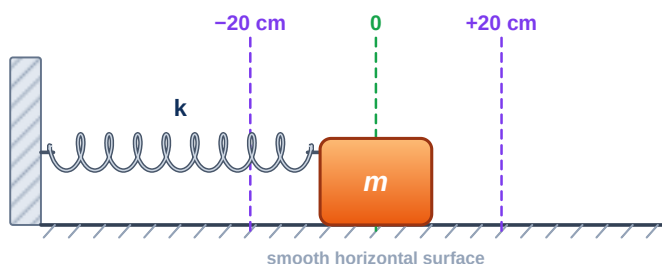
I The periodic time of a mass-spring oscillator does not depend on the amplitude.

II The speed of the oscillating body is maximum at the equilibrium position.

III A horizontal mass-spring oscillator has the same periodic time on the Moon as on the Earth.

27

A body of mass 1 kg oscillates on a horizontal spring of constant 100 N/m with an amplitude of 20 cm:





Stand Still, Keep Vibrating



- I** Its speed at a displacement of 12 cm equals
- a) 2 m/s b) 1.6 m/s c) 1.2 m/s d) 0.8 m/s

- II** Find the kinetic and potential energies at that point, and check that their sum is constant.

28

The displacement of a body is $x = 6 \sin(\pi t)$, where x is in cm and t in seconds:

- I** The first moment at which $x = 3$ cm is
- a) 0.5 s b) $1/12$ s c) $1/3$ s d) $1/6$ s

- II** Find the speed of the body at that moment.

29

A body in simple harmonic motion has an acceleration of magnitude 8 m/s^2 when its displacement is 2 cm:

- I** Its angular frequency equals
- a) 400 rad/s b) 4 rad/s c) 2 rad/s d) 20 rad/s

- II** At what displacement is the magnitude of its acceleration 12 m/s^2 ?

- III** Find the periodic time.

30

Two bodies, each of mass 1 kg, oscillate on two springs of constants $k_1 = 100 \text{ N/m}$ and $k_2 = 400 \text{ N/m}$, both with the same amplitude of 5 cm:

- I** The ratio of their frequencies f_1 / f_2 equals
- a) 2 b) 4 c) $1/4$ d) $1/2$

I Find the ratio of their total energies E_1 / E_2 .

III Find the ratio of their maximum speeds.

31

A body oscillates in simple harmonic motion with an amplitude of 10 cm and a periodic time of 2.4 s:

I The shortest time the body takes to move directly from $x = +5$ cm to $x = -5$ cm is

a) 0.2 s b) 0.8 s c) 0.4 s d) 1.2 s

I Find the average speed of the body during this interval, and compare it with its maximum speed.

III During one complete oscillation, for how long is the body more than 5 cm away from the equilibrium position?

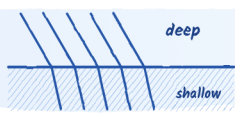
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When the displacement of a body in simple harmonic motion is 3 cm, the magnitude of its acceleration is 12 m/s^2 and its speed is 0.8 m/s:

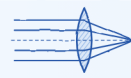
I The angular frequency of the motion is

a) 400 rad/s b) 20 rad/s c) 4 rad/s d) 26.7 rad/s

II Find the amplitude and the maximum speed.



Enter
Bend
Continue



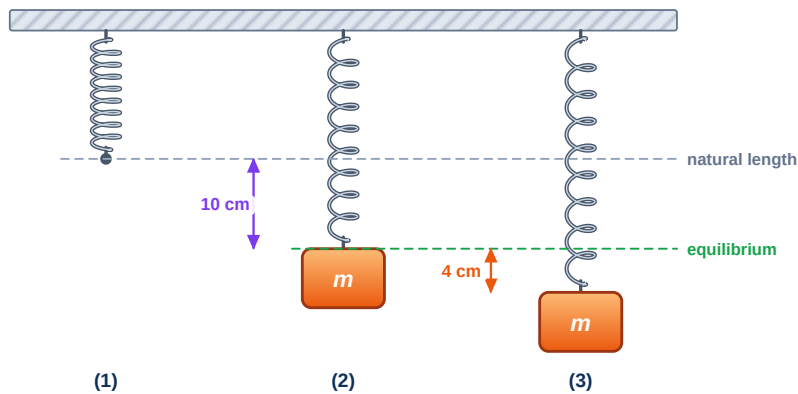
Bend,
Don't
Break



III The body is moving away from the equilibrium position. How long does it take to reach the extreme position?

33

A light spring hangs vertically. When a body of mass 0.4 kg is hung from it, the spring stretches 10 cm and the body rests at its equilibrium position. The body is then pulled 4 cm further down and released ($g = 10 \text{ m/s}^2$):



I The spring constant equals

a) 4 N/m

b) 400 N/m

c) 0.04 N/m

d) 40 N/m

II The periodic time of the oscillation equals

a) 0.628 s

b) 0.2 s

c) 6.28 s

d) 0.0628 s

III Find the greatest and the smallest tension in the spring during the motion.

IV What is the largest amplitude for which the spring is never compressed during the motion?



Vibrate
Travel
Hear

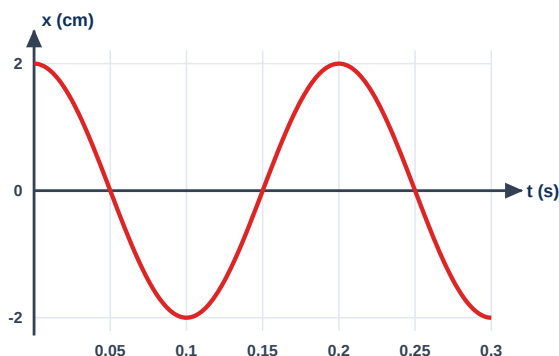


Sound
Mind,
Sound Physics



34

The graph shows the displacement of a body of mass 0.5 kg attached to a horizontal spring against time:



I The spring constant equals

a) 157 N/m

b) 98.6 N/m

c) 493 N/m

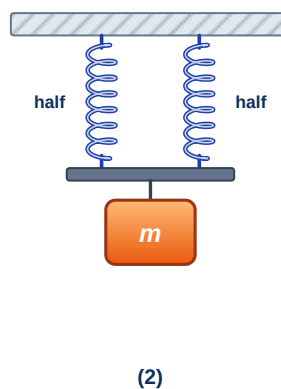
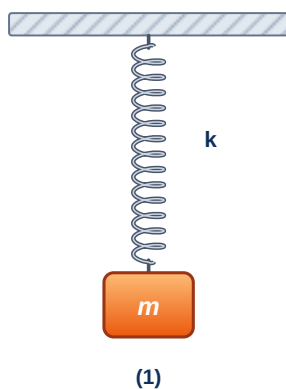
d) 4930 N/m

II Find the speed of the body when its displacement is 1 cm .

III Find the first moment after $t = 0$ at which the kinetic energy equals the potential energy.

35

A body hung from a spring oscillates vertically with a periodic time of 0.8 s . The spring is then cut into two equal halves, as shown:





Vibrate
Travel
Hear



Z BANK

Oscillations & Waves · Question Bank



Sound
Mind,
Sound Physics



- I** The spring constant of each half is
 a) $k/2$ b) $2k$ c) k d) $4k$
- I** If the body is hung from one half only, its periodic time becomes
 a) 0.566 s b) 0.4 s c) 1.13 s d) 0.8 s
- II** The body is hung from the two halves side by side, as in figure (2). Find its periodic time.

36

A block of mass 0.5 kg on a smooth horizontal surface oscillates on a spring of constant 50 N/m with an amplitude of 8 cm . At the moment the block reaches an extreme position, a second block of mass 0.5 kg is placed gently on it and moves with it:

- I** The amplitude of the new motion is
 a) 4 cm b) 5.66 cm c) 11.3 cm d) 8 cm
- I** Find the periodic time before and after adding the block.
- II** Find the maximum speed before and after, and show that the total energy has not changed.

37

The displacement of a body in simple harmonic motion is $x = 0.04 \sin(10\pi t)$, where x is in metres and t in seconds:

- I** The first moment at which the speed of the body is half its maximum speed is
 a) $1/60\text{ s}$ b) $1/15\text{ s}$ c) $1/20\text{ s}$ d) $1/30\text{ s}$

I Find the second moment at which this happens, and the acceleration of the body at these two moments.

III Find the distance travelled by the body from $t = 0$ to $t = 0.25$ s.



A body of mass 0.5 kg moves in simple harmonic motion. Its kinetic energy is 0.27 J when its displacement is 3 cm, and 0.11 J when its displacement is 5 cm:

I The spring constant equals

a) 100 N/m

b) 200 N/m

c) 400 N/m

d) 80 N/m

II Find the total energy and the amplitude.

III Find the maximum speed and the periodic time.



Two mass-spring oscillators A and B have the same total energy. The mass of A is twice the mass of B, and the spring constant of A is 4 times that of B:

I The ratio of their amplitudes A_A / A_B equals

a) 2

b) 1/2

c) 1/4

d) $1/\sqrt{2}$

II The ratio of their maximum speeds v_A / v_B equals

a) $\sqrt{2}$

b) 1/2

c) 2

d) $1/\sqrt{2}$

III Find the ratio of their periodic times and the ratio of their maximum accelerations.

40

A body of mass 0.3 kg on a spring starts its simple harmonic motion from the equilibrium position ($x = A \sin \omega t$). It is 2 cm away from the equilibrium position for the first time at $t = 0.05$ s, and it reaches an extreme position for the first time at $t = 0.15$ s:

I The periodic time of the motion is

a) 0.15 s

b) 0.3 s

c) 0.6 s

d) 1.2 s

II The amplitude of the motion is

a) 4 cm

b) 2 cm

c) 2.83 cm

d) 8 cm

III Find the spring constant and the total energy of the oscillator.

Chapter 2 · Lesson 1

Model Answers

Every question solved step by step

1

I $\omega = \sqrt{k / m} = \sqrt{50 / 0.5} = \sqrt{100}$

$\omega = 10 \text{ rad/s}$

d) 10 rad/s

II $T = 2\pi / \omega = (2 \times 3.14) / 10$

$T = 0.628 \text{ s}$

a) 0.628 s

2

I Comparing with $x = A \sin(\omega t)$:
 $A = 0.05 \text{ m}$

b) 0.05 m

II $\omega = 20 \text{ rad/s}$
 $f = \omega / 2\pi = 20 / 6.28 = 3.18 \text{ Hz}$

c) 3.18 Hz

3

I $F = -k x$: proportional to the displacement x ,
and the negative sign means it points towards $x = 0$.

b) directed towards the equilibrium position and proportional to the displacement

II At $x = 0$ all the energy is kinetic, so the speed is maximum : $v_{\text{max}} = \omega A$.

a) the equilibrium position

III $a = -\omega^2 x$: its magnitude is largest where $|x|$ is largest, at $x = \pm A$.

c) the two extreme positions

4

I The largest displacement read from the graph is 4 cm.
 $A = 4 \text{ cm}$

b) 4 cm

II One complete cycle takes $T = 0.8 \text{ s}$
 $f = 1 / T = 1 / 0.8 = 1.25 \text{ Hz}$

c) 1.25 Hz

III $\omega = 2\pi / T = 6.28 / 0.8 = 7.85 \text{ rad/s}$

$\omega = 7.85 \text{ rad/s}$

5

I $F = -k x \rightarrow k = |F| / x$
 $k = 20 / 0.1 = 200 \text{ N/m}$

c) 200 N/m

II $T = 2\pi \sqrt{m / k} = 6.28 \times \sqrt{2 / 200}$
 $T = 6.28 \times 0.1 = 0.628 \text{ s}$

a) 0.628 s

- III The restoring force is always opposite to the displacement ($F = -kx$): it always pulls the body back towards the equilibrium position.

F is always opposite to x

6

I $a_{\max} / v_{\max} = \omega^2 A / \omega A = \omega$
 $\omega = 40 / 2 = 20 \text{ rad/s}$

b) 20 rad/s

II $A = v_{\max} / \omega = 2 / 20 = 0.1 \text{ m}$
 $T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$

A = 0.1 m, T = 0.314 s

7

I $T \propto 1 / \sqrt{k} \rightarrow T_A / T_B = \sqrt{4k / k} = 2$

d) 2

II $T_B = 2\pi \sqrt{M / 4k}$ must equal $T_A = 2\pi \sqrt{m / k}$
 $M / 4k = m / k \rightarrow M = 4m$

M = 4m

8

I $E = \frac{1}{2} k A^2 = \frac{1}{2} \times 80 \times (0.1)^2$
 $E = 0.4 \text{ J}$

b) 0.4 J

II At $x = 0$ all the energy is kinetic: $\frac{1}{2} m v_{\max}^2 = E$
 $v_{\max} = \sqrt{2E / m} = \sqrt{2 \times 0.4 / 0.2} = 2 \text{ m/s}$

$v_{\max} = 2 \text{ m/s}$

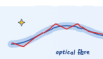
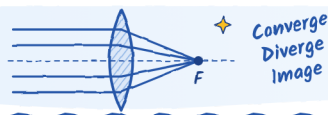
III $PE = \frac{1}{2} k x^2 = \frac{1}{2} \times 80 \times (0.05)^2 = 0.1 \text{ J}$
 $KE = E - PE = 0.4 - 0.1 = 0.3 \text{ J}$
 $v = \sqrt{2 KE / m} = \sqrt{2 \times 0.3 / 0.2} = \sqrt{3} \approx 1.73 \text{ m/s}$

$v \approx 1.73 \text{ m/s}$

9

- I The kinetic energy is maximum at $x = 0$ and zero at $x = \pm A \rightarrow$ curve (2).
 Curve (1) is the potential energy $\frac{1}{2} kx^2$, zero at $x = 0$.

d) curve (2)



I $PE = KE = E/2 \rightarrow \frac{1}{2} k x^2 = \frac{1}{2} (\frac{1}{2} k A^2)$
 $x^2 = A^2 / 2 \rightarrow x = A / \sqrt{2} \approx 0.71 A$

c) $A / \sqrt{2}$

III $PE = \frac{1}{2} k (A/2)^2 = E / 4$
 $KE = E - E/4 = 3E / 4$
 $KE : PE = 3 : 1$

KE : PE = 3 : 1

10

I $A/2 = A \sin(\omega t) \rightarrow \sin(\omega t) = 0.5 \rightarrow \omega t = \pi/6$
 $t = (\pi/6) / (2\pi/T) = T / 12 = 1.2 / 12 = 0.1 \text{ s}$

c) 0.1 s

II from 0 to A takes $T/4 = 0.3 \text{ s}$
from A/2 to A : $0.3 - 0.1 = 0.2 \text{ s}$
the body slows down as it approaches the extreme position, so the second half takes longer

t = 0.2 s

11

I $a = -\omega^2 x \rightarrow \omega^2 = |a| / x = 8 / 0.02 = 400$
 $\omega = 20 \text{ rad/s}$

d) 20 rad/s

II $T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$

T = 0.314 s

III The acceleration is proportional to the displacement and always opposite to it ($a = -\omega^2 x$).

a $\propto$ -x

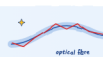
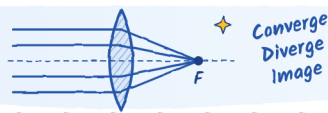
12

I $x = 0.1 \sin(5\pi \times 0.1) = 0.1 \sin(\pi/2)$
 $x = 0.1 \times 1 = 0.1 \text{ m}$ (an extreme position)

c) 0.1 m

II At an extreme position the body stops for an instant : $v = 0$
 $a = -\omega^2 x = -(5 \times 3.14)^2 \times 0.1 = -246.5 \times 0.1 \approx -24.6 \text{ m/s}^2$

v = 0 , a $\approx$ -24.6 m/s²



III $\omega = 5\pi \rightarrow T = 2\pi / \omega = 2\pi / 5\pi = 0.4 \text{ s}$

$T = 0.4 \text{ s}$

13

- I The periodic time of simple harmonic motion does not depend on the amplitude; that is why the pitch of the sound stays the same.

c) remains constant

- II A shorter free length is stiffer (a larger k),
 $f = (1 / 2\pi) \sqrt{k / m} \rightarrow$ the frequency increases (a higher pitch).

the frequency increases

14

- I Both take $T/4$, and $T = 2\pi \sqrt{m / k}$ does not depend on the amplitude.

c) both at the same time

- II $v_{\max} = \omega A$, with the same ω
 $v_A : v_B = 2 : 5$

2 : 5

- III $E = \frac{1}{2} k A^2 \rightarrow E \propto A^2$
 $E_A : E_B = 2^2 : 5^2 = 4 : 25$

4 : 25

15

- I $T = 2\pi \sqrt{m/k}$: unchanged

$v_{\max} = \omega A$: doubled

$E = \frac{1}{2} k A^2$: $\times 4$

$a_{\max} = \omega^2 A$: doubled

b) the maximum speed is doubled

- II $E \propto A^2 \rightarrow E' / E = 2^2 = 4$

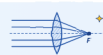
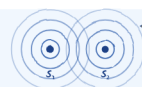
c) 4 times

16

I $T' / T = \sqrt{m' / m} \times \sqrt{k / k'} = \sqrt{2} \times \sqrt{1/4}$

$T' / T = \sqrt{2} / 2 = 1 / \sqrt{2}$

a) $T / \sqrt{2}$



I $f \propto 1/T \rightarrow f' = \sqrt{2} f \approx 1.41 f$

$f' = \sqrt{2} f$

17

I The speed is maximum where $x = 0$ (the steepest parts of the curve) : P, R and T.

a) P, R and T

II $a = -\omega^2 x$ is largest and positive where $x = -A$: point S.

d) S

III $F = -kx = 0$ where $x = 0$: P, R and T.

P, R and T

18

I energy : $\frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2 \rightarrow v = \omega \sqrt{A^2 - x^2}$

$$v = 10 \times \sqrt{0.01 - 0.0036} = 10 \times 0.08 = 0.8 \text{ m/s}$$

c) 0.8 m/s

II $v = v_{\max} / 2 \rightarrow \omega \sqrt{A^2 - x^2} = \omega A / 2$

$$A^2 - x^2 = A^2 / 4 \rightarrow x = (\sqrt{3} / 2) A$$

$$x = 0.866 \times 10 = 8.66 \text{ cm}$$

$x \approx 8.66 \text{ cm}$

19

I $v^2 = \omega^2 (A^2 - x^2)$ for both positions

$$0.64 = \omega^2 (A^2 - 0.0036) \text{ and } 0.36 = \omega^2 (A^2 - 0.0064)$$

$$\text{subtracting : } 0.28 = \omega^2 \times 0.0028 \rightarrow \omega^2 = 100 \rightarrow \omega = 10 \text{ rad/s}$$

$\omega = 10 \text{ rad/s}$

II $A^2 = 0.64 / 100 + 0.0036 = 0.0064 + 0.0036 = 0.01$

$$A = 0.1 \text{ m} = 10 \text{ cm}$$

$A = 10 \text{ cm}$

20

I $T = 0.2 \text{ s}$ (one complete cycle)

$$\omega = 2\pi / T = 6.28 / 0.2 = 31.4 \text{ rad/s}$$

$\omega = 31.4 \text{ rad/s}$

I $v_{\max} = \omega A \rightarrow A = 0.5 / 31.4 = 0.0159 \text{ m} \approx 1.59 \text{ cm}$

a) 1.59 cm

III At $t = 0$ the speed is maximum, so the body is at the equilibrium position.

at the equilibrium position

21

I $PE = \frac{1}{2} k (A/3)^2 = E / 9$

$KE = E - E/9 = 8E / 9$

$KE : PE = 8$

b) 8

II $\omega = 2\pi f = 31.4 \text{ rad/s}$

$F_{\max} = m \omega^2 A = 0.5 \times 986 \times 0.04 \approx 19.7 \text{ N}$

a) 19.7 N

22

I $T^2 \propto m \rightarrow (0.7 / 0.5)^2 = (m_1 + 0.3) / m_1$

$1.96 m_1 = m_1 + 0.3$

$0.96 m_1 = 0.3 \rightarrow m_1 \approx 0.31 \text{ kg}$

$m_1 \approx 0.31 \text{ kg}$

II $k = 4\pi^2 m_1 / T^2 = 39.44 \times 0.3125 / 0.25$

$k \approx 49.3 \text{ N/m}$

$k \approx 49.3 \text{ N/m}$

23

I $T^2 = (4\pi^2 / k) m \rightarrow \text{slope} = 4\pi^2 / k$

$\text{slope} = 0.4 / 0.5 = 0.8 \text{ s}^2/\text{kg}$

$k = 4\pi^2 / 0.8 = 39.44 / 0.8 \approx 49.3 \text{ N/m}$

a) 49.3 N/m

II $T^2 = 0.8 \times 0.8 = 0.64 \text{ s}^2$

$T = 0.8 \text{ s}$

$T = 0.8 \text{ s}$

**24**

I $T = 0.5 \text{ s}$
 $\omega = 2\pi / T = 6.28 / 0.5 = 12.56 \text{ rad/s}$

c) 12.56 rad/s

II $A = 3 \text{ cm} = 0.03 \text{ m}$, $\omega = 12.56 \text{ rad/s}$

$x = 0.03 \sin (12.56 t)$

25

I $x = A \sin (2\pi/T \times T/8) = A \sin 45^\circ = A / \sqrt{2}$
 $PE = \frac{1}{2} k A^2 / 2 = E / 2 \rightarrow KE = E / 2$
 $KE : PE = 1 : 1$

a) $1 : 1$

II $\omega^2 = a_{\max} / A = 10 / 0.025 = 400 \rightarrow \omega = 20 \text{ rad/s}$
 $f = \omega / 2\pi = 20 / 6.28 = 3.18 \text{ Hz}$

a) 3.18 Hz

26

I A larger amplitude gives a larger restoring force and a larger acceleration, so the longer path is covered faster in the same time : $T = 2\pi \sqrt{m/k}$ contains no A .

T depends only on m and k

II There the potential energy is zero, so all the energy is kinetic.

$PE = 0 \rightarrow KE$ is maximum

III $T = 2\pi \sqrt{m/k}$ does not contain g ; the mass and the spring do not change.

T does not depend on g

27

I $\omega = \sqrt{100 / 1} = 10 \text{ rad/s}$
 $v = \omega \sqrt{A^2 - x^2} = 10 \times \sqrt{0.04 - 0.0144} = 10 \times 0.16 = 1.6 \text{ m/s}$

b) 1.6 m/s

II $KE = \frac{1}{2} m v^2 = \frac{1}{2} \times 1 \times (1.6)^2 = 1.28 \text{ J}$
 $PE = \frac{1}{2} k x^2 = \frac{1}{2} \times 100 \times (0.12)^2 = 0.72 \text{ J}$
 $KE + PE = 2 \text{ J} = \frac{1}{2} k A^2 = \frac{1}{2} \times 100 \times (0.2)^2 \checkmark$

$KE = 1.28 \text{ J}$, $PE = 0.72 \text{ J}$



Strike
Listen
Tune



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Oscillations & Waves · Question Bank



Tune
In To
Physics



28

I $3 = 6 \sin(\pi t) \rightarrow \sin(\pi t) = 0.5 \rightarrow \pi t = \pi/6$
 $t = 1/6 \text{ s} \approx 0.167 \text{ s}$

d) $1/6 \text{ s}$

II $v = \omega \sqrt{A^2 - x^2} = \pi \sqrt{36 - 9} = 3.14 \times 5.2$
 $v \approx 16.3 \text{ cm/s}$

v $\approx 16.3 \text{ cm/s}$

29

I $|a| = \omega^2 x \rightarrow \omega^2 = 8 / 0.02 = 400$
 $\omega = 20 \text{ rad/s}$

d) 20 rad/s

II $x = |a| / \omega^2 = 12 / 400 = 0.03 \text{ m}$

x = 3 cm

III $T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$

T = 0.314 s

30

I $\omega_1 = \sqrt{100 / 1} = 10 \text{ rad/s}$, $\omega_2 = \sqrt{400 / 1} = 20 \text{ rad/s}$
 $f_1 / f_2 = \omega_1 / \omega_2 = 1/2$

d) $1/2$

II $E = \frac{1}{2} k A^2$, with the same A $\rightarrow E \propto k$
 $E_1 / E_2 = 100 / 400 = 1/4$
 $(E_1 = 0.125 \text{ J}, E_2 = 0.5 \text{ J})$

$E_1 / E_2 = 1/4$

III $v_{\max} = \omega A \rightarrow v_1 / v_2 = 10 / 20 = 1/2$
 $(v_1 = 0.5 \text{ m/s}, v_2 = 1 \text{ m/s})$

$1/2$

31

I from $x = 0$ to $x = A/2$: $A/2 = A \sin(\omega t) \rightarrow \sin(\omega t) = 1/2 \rightarrow \omega t = \pi/6$
 $t = (\pi/6) / (2\pi/T) = T / 12 = 2.4 / 12 = 0.2 \text{ s}$
 from $+A/2$ to 0 takes 0.2 s and from 0 to $-A/2$ takes another 0.2 s
 $t = 0.2 + 0.2 = 0.4 \text{ s}$

c) 0.4 s

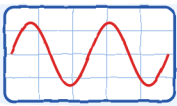
26

Stay Tuned ... Keep Solving

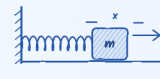
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Oscillations & Waves

CHAPTER
2



Measure
Analyse
Solve



Frequency
Of
Success



- I** average speed = distance / time = $0.1 / 0.4 = 0.25 \text{ m/s}$
 $v_{\max} = \omega A = (2\pi / T) A = (6.28 / 2.4) \times 0.1 \approx 0.262 \text{ m/s}$
the average is less than $v_{\max}$ because the speed is maximum only at $x = 0$
average = 0.25 m/s , $v_{\max} \approx 0.262 \text{ m/s}$

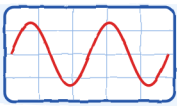
- III** in one oscillation the body crosses the region $-A/2 < x < +A/2$ twice,
and each crossing takes $2 \times T/12 = T/6$
time inside the region = $2 \times T/6 = T/3 = 0.8 \text{ s}$
time outside = $T - T/3 = 2.4 - 0.8 = 1.6 \text{ s}$
 $t = 1.6 \text{ s}$

32

- I** $|a| = \omega^2 x \rightarrow \omega^2 = 12 / 0.03 = 400$
 $\omega = 20 \text{ rad/s}$
b) 20 rad/s
- II** $v = \omega \sqrt{A^2 - x^2} \rightarrow 0.8 = 20 \sqrt{A^2 - 0.0009}$
 $\sqrt{A^2 - 0.0009} = 0.04 \rightarrow A^2 = 0.0016 + 0.0009 = 0.0025$
 $A = 0.05 \text{ m} = 5 \text{ cm}$
 $v_{\max} = \omega A = 20 \times 0.05 = 1 \text{ m/s}$
 $A = 5 \text{ cm}$, $v_{\max} = 1 \text{ m/s}$
- III** $T = 2\pi / \omega = 6.28 / 20 = 0.314 \text{ s}$
 $x = A \sin \theta \rightarrow \sin \theta = 3 / 5 = 0.6 \rightarrow \theta = 36.87^\circ$
the extreme position is at $\theta = 90^\circ$: the remaining angle is $90^\circ - 36.87^\circ = 53.13^\circ$
 $t = (53.13 / 360) \times 0.314 \approx 0.046 \text{ s}$
 $t \approx 0.046 \text{ s}$

33

- I** at equilibrium the spring force balances the weight : $k e = m g$
 $k = m g / e = (0.4 \times 10) / 0.1 = 40 \text{ N/m}$
d) 40 N/m
- II** $T = 2\pi \sqrt{m / k} = 6.28 \times \sqrt{0.4 / 40} = 6.28 \times 0.1 = 0.628 \text{ s}$
(the same as $2\pi \sqrt{e / g} = 6.28 \times \sqrt{0.1 / 10}$)
a) 0.628 s
- III** the amplitude is 4 cm about the equilibrium position (extension 10 cm)
lowest point : extension = $10 + 4 = 14 \text{ cm} \rightarrow T_{\max} = 40 \times 0.14 = 5.6 \text{ N}$
highest point : extension = $10 - 4 = 6 \text{ cm} \rightarrow T_{\min} = 40 \times 0.06 = 2.4 \text{ N}$
 $T_{\max} = 5.6 \text{ N}$, $T_{\min} = 2.4 \text{ N}$

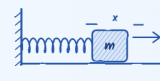


Measure
Analyse
Solve



Oscillations & Waves · Question Bank

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Frequency
Of
Success



- IV** at the highest point the extension is 10 cm – A
the spring is never compressed if $10 - A \geq 0$
 $A \leq 10$ cm (the static extension)

$A_{\max} = 10 \text{ cm}$

34

- I** from the graph : $A = 2 \text{ cm}$, $T = 0.2 \text{ s}$
 $\omega = 2\pi / T = 6.28 / 0.2 = 31.4 \text{ rad/s}$
 $k = m \omega^2 = 0.5 \times (31.4)^2 = 0.5 \times 986 \approx 493 \text{ N/m}$

c) 493 N/m

- II** $v = \omega \sqrt{A^2 - x^2} = 31.4 \times \sqrt{0.02^2 - 0.01^2}$
 $v = 31.4 \times \sqrt{0.0003} = 31.4 \times 0.0173 \approx 0.544 \text{ m/s}$

v $\approx$ 0.544 m/s

- III** $KE = PE = E/2 \rightarrow \frac{1}{2} k x^2 = \frac{1}{4} k A^2 \rightarrow x = A / \sqrt{2}$
the body starts at $x = A : x = A \cos(\omega t) \rightarrow \cos(\omega t) = 1 / \sqrt{2} \rightarrow \omega t = \pi/4$
 $t = (\pi/4) / (2\pi/T) = T / 8 = 0.2 / 8 = 0.025 \text{ s}$

t = 0.025 s

35

- I** the same force F stretches the whole spring x and each half only $x/2$
 $k_{\text{half}} = F / (x/2) = 2 F / x = 2k$

b) 2k

- II** $T \propto 1 / \sqrt{k} \rightarrow T' = T / \sqrt{2} = 0.8 / 1.414 \approx 0.566 \text{ s}$

a) 0.566 s

- III** both halves stretch the same distance x
total force = $2k x + 2k x = 4k x \rightarrow$ the pair acts as one spring $4k$
 $T'' = T / \sqrt{4} = 0.8 / 2 = 0.4 \text{ s}$

T'' = 0.4 s

36

- I** at the extreme position the speed is zero, and adding the block there gives it no speed
so the body is still at rest at 8 cm from equilibrium : $A = 8 \text{ cm}$

d) 8 cm

I before : $T_1 = 2\pi \sqrt{0.5 / 50} = 6.28 \times 0.1 = 0.628 \text{ s}$

after : $T_2 = 2\pi \sqrt{1 / 50} = 6.28 \times 0.141 \approx 0.888 \text{ s}$

$T_1 = 0.628 \text{ s}, T_2 \approx 0.888 \text{ s}$

III $\omega_1 = \sqrt{50 / 0.5} = 10 \text{ rad/s} \rightarrow v_1 = 10 \times 0.08 = 0.8 \text{ m/s}$

$\omega_2 = \sqrt{50 / 1} \approx 7.07 \text{ rad/s} \rightarrow v_2 = 7.07 \times 0.08 \approx 0.566 \text{ m/s}$

$E = \frac{1}{2} k A^2 = \frac{1}{2} \times 50 \times (0.08)^2 = 0.16 \text{ J}$ in both cases (k and A did not change)

check : $\frac{1}{2} \times 1 \times (0.566)^2 \approx 0.16 \text{ J}$

$v_1 = 0.8 \text{ m/s}, v_2 \approx 0.566 \text{ m/s}, E = 0.16 \text{ J}$

37

I $v = v_{\max} / 2 \rightarrow \omega \sqrt{A^2 - x^2} = \omega A / 2 \rightarrow x = (\sqrt{3} / 2) A$

$\sin(10\pi t) = \sqrt{3} / 2 \rightarrow 10\pi t = \pi/3$

$t = 1/30 \text{ s}$

d) $1/30 \text{ s}$

II $\sin(10\pi t) = \sqrt{3} / 2$ again when $10\pi t = 2\pi/3 \rightarrow t = 1/15 \text{ s}$

at both moments $x = (\sqrt{3} / 2) \times 0.04 \approx 0.0346 \text{ m}$

$a = -\omega^2 x = -(10 \times 3.14)^2 \times 0.0346 = -986 \times 0.0346 \approx -34.2 \text{ m/s}^2$

$t = 1/15 \text{ s}, a \approx -34.2 \text{ m/s}^2$

III $\omega = 10\pi \rightarrow T = 2\pi / \omega = 0.2 \text{ s}$

$0.25 \text{ s} = T + T/4$

one oscillation covers 4A and a quarter oscillation (from $x = 0$ to $x = A$) covers A
distance = $4A + A = 5A = 5 \times 4 = 20 \text{ cm}$

distance = 20 cm

38

I the total energy is constant : $KE_1 + \frac{1}{2} k x_1^2 = KE_2 + \frac{1}{2} k x_2^2$

$0.27 - 0.11 = \frac{1}{2} k (0.05^2 - 0.03^2) = \frac{1}{2} k \times 0.0016$

$0.16 = 0.0008 k \rightarrow k = 200 \text{ N/m}$

b) 200 N/m

II $E = KE_1 + \frac{1}{2} k x_1^2 = 0.27 + \frac{1}{2} \times 200 \times 0.0009 = 0.27 + 0.09 = 0.36 \text{ J}$

$E = \frac{1}{2} k A^2 \rightarrow A^2 = 2 \times 0.36 / 200 = 0.0036 \rightarrow A = 0.06 \text{ m}$

$E = 0.36 \text{ J}, A = 6 \text{ cm}$

III $\frac{1}{2} m v_{\max}^2 = E \rightarrow v_{\max} = \sqrt{2 \times 0.36 / 0.5} = \sqrt{1.44} = 1.2 \text{ m/s}$

$\omega = \sqrt{k / m} = \sqrt{200 / 0.5} = 20 \text{ rad/s} \rightarrow T = 6.28 / 20 = 0.314 \text{ s}$

check : $\omega A = 20 \times 0.06 = 1.2 \text{ m/s}$

$v_{\max} = 1.2 \text{ m/s}, T = 0.314 \text{ s}$

39

I $E = \frac{1}{2} k A^2$ is the same $\rightarrow k_A A_A^2 = k_B A_B^2$
 $A_A / A_B = \sqrt{k_B / k_A} = \sqrt{1/4} = 1/2$

b) $1/2$

II $E = \frac{1}{2} m v_{\max}^2$ is the same $\rightarrow v_A / v_B = \sqrt{m_B / m_A}$
 $v_A / v_B = \sqrt{1/2} = 1/\sqrt{2}$

d) $1/\sqrt{2}$

III $T = 2\pi \sqrt{m / k} \rightarrow T_A / T_B = \sqrt{(m_A / m_B) \times (k_B / k_A)} = \sqrt{2 \times 1/4} = 1/\sqrt{2}$
 $a_{\max} = \omega^2 A = (k / m) A$
 $a_A / a_B = (k_A / k_B) \times (m_B / m_A) \times (A_A / A_B) = 4 \times 1/2 \times 1/2 = 1$

$T_A / T_B = 1/\sqrt{2}, a_A / a_B = 1$

40

I from the equilibrium position to an extreme position takes a quarter of a cycle
 $T / 4 = 0.15 \text{ s} \rightarrow T = 0.6 \text{ s}$

c) 0.6 s

II $\omega t = (2\pi / 0.6) \times 0.05 = \pi/6$
 $x = A \sin(\pi/6) \rightarrow 2 = A \times 0.5 \rightarrow A = 4 \text{ cm}$

a) 4 cm

III $\omega = 2\pi / T = 6.28 / 0.6 \approx 10.47 \text{ rad/s}$
 $k = m \omega^2 = 0.3 \times (10.47)^2 = 0.3 \times 109.6 \approx 32.9 \text{ N/m}$
 $E = \frac{1}{2} k A^2 = \frac{1}{2} \times 32.9 \times (0.04)^2 \approx 0.0263 \text{ J}$

$k \approx 32.9 \text{ N/m}, E \approx 0.0263 \text{ J}$